

Mar 3 2026

Ex. $\dot{x} = -\frac{x}{1+t}$

$\dot{x}(t) = A(t)x$

$x(t) = x(t_0) \frac{1+t_0}{1+t}$

$t_0 > -1$

$\dot{x} = f(x, t)$

Definition: $x=0$ is stable if for every $\varepsilon > 0$ and t_0 there exists $\delta > 0$ s.t.

$|x(t_0)| \leq \delta(t_0, \varepsilon) \Rightarrow |x(t)| < \varepsilon \quad \forall t \geq t_0$

If the same δ works for all t_0

i.e. $\delta = \delta(\varepsilon)$ the $x=0$ is uniformly stable

$\varepsilon = 1$

$t_0 = 0 \quad |x(t_0)| \leq 1 \leftarrow \delta$

$t_0 = 5 \quad |x(t_0)| \leq 5 \leftarrow \delta = 1$

$t_0 = 10 \quad |x(t_0)| \leq 2 \leftarrow \delta$

$\dot{x} = A(t)x$

$A(t) = \begin{bmatrix} -(1 + 1.5 \cos^2 t) & (-1.5 \sin t \cos t) \\ -(-1 - 1.5 \sin t \cos t) & -1 + 1.5 \sin^2 t \end{bmatrix}$

e-values: $-0.25 \mp j 0.25 \sqrt{7}$

$x(t) = \underbrace{\begin{bmatrix} e^{0.5t} \cos t & e^{-t} \sin t \\ e^{-0.5t} \sin t & e^{-t} \cos t \end{bmatrix}}_{\Phi(t, 0)} x(0)$

, If $w_1(x) \leq v(t, x) \leq w_2(x)$

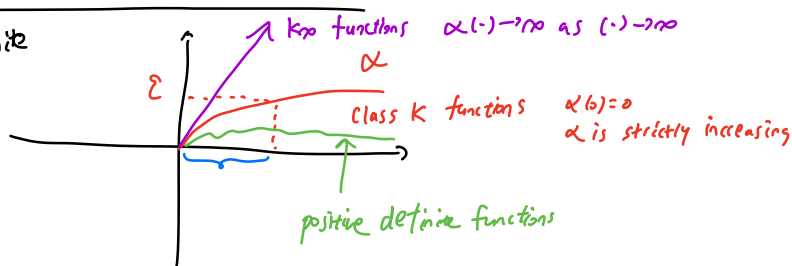
and $\dot{v}(t, x) \triangleq \frac{\partial v}{\partial t} + \frac{\partial v}{\partial x} f(t, x) \leq 0$

for some positive definite w_1 and w_2

then $x=0$ is uniformly stable

stability definitions through comparison functions

$K_{\infty} \subset K \subset \text{Positive definite}$



$x=0$ is uniformly stable if:

$$|x(t)| \leq \alpha(|x(t_0)|) \quad \text{for } |x(t_0)| < c \quad \text{for some } c > 0$$

$x=0$ is uniformly asymptotically stable

if there exists a class-KL function $\beta(\cdot, \cdot)$

$$\text{s.t. } |x(t)| \leq \beta(|x(t_0)|, t-t_0)$$

for $|x(t_0)| < c$ for some $c > 0$

$\beta(\cdot, \cdot)$ is class-KL if:

1. $\beta(\cdot, s)$ is class-K for every fixed s
2. $\beta(r, \cdot)$ is decreasing and $\beta(r, s) \rightarrow 0$ as $s \rightarrow \infty$ for every fixed r

- If $w_1(x) \leq v(t, x) \leq w_2(x)$
and $\dot{v}(t, x) \leq -w_3(x)$ for some positive definite $w_3(\cdot)$
then $x=0$ is uniformly asymptotically stable

- $[+]$ $w_1(x)$ is radially unbounded

$\Rightarrow x=0$ is uniformly GAS

Ex. $\dot{x} = -x^3$ (Recall)

$$V(x) = \frac{1}{2}x^2$$

$$\dot{V}(x) = x(-x^3) = -x^4 < 0, x \neq 0$$

radially unbounded

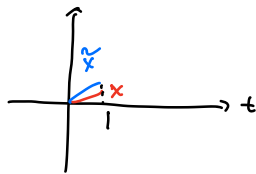
$\Rightarrow x=0$ is GAS

Comparison Lemma

$$x \in \mathbb{R} \quad \dot{x} = f(x, t) \leq g(x) \quad \forall t; x$$

$$\dot{\tilde{x}} = f(\tilde{x}, t)$$

$$\begin{aligned} \dot{\tilde{x}} &= g(\tilde{x}) & x(t_0) &\leq \tilde{x}(t_0) \\ & & \Downarrow & \\ x(t) &\leq \tilde{x}(t) & \forall t \end{aligned}$$



Ex.

$$\begin{aligned} \dot{x} &= \overbrace{-h(t)x^3}^{f(x,t)} & h(t) &\geq 1 \\ &\leq -x^3 = g(x) \end{aligned}$$

$$V(t, x) \equiv V(x) = \frac{1}{2} x^2$$

w_1, w_2 : Positive Definite

$$w_1 = w_2 = \frac{1}{2} x^2$$

$$\dot{V} = x(-h(t)x^3)$$

$$= -h(t)x^4 \leq -w_3(x)$$

$$w_3 = \frac{1}{2} x^4 \text{ positive definite}$$

$\Rightarrow x=0$ is uniformly asymptotically stable

Ex.

$$\dot{x} = \frac{-x}{1+t}$$

$$x(t) = \frac{x(t_0)}{1 + \frac{t-t_0}{1+t_0}}$$

$$|x(t)| \leq |x(t_0)| \underbrace{\frac{1}{1 + \frac{t-t_0}{1+t_0}}}_{\text{not a class KL function}}$$

not a class KL function

